

Curriculum Vitae

Michael Peter Matessa

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Education

Ph.D. Cognitive Psychology, Carnegie Mellon University, 2000
Dissertation title: "Simulating Adaptive Communication"

M.S. Psychology, Carnegie Mellon University, 1998
Thesis title: "Modeling Focused Learning in Role Assignment"

B.S. Carnegie Mellon University 1989
Double major: Applied Mathematics (Computer Science) and Cognitive Science

Professional Experience

Principal Systems Engineer/ Collins Aerospace (formerly Rockwell Collins) 2012-2025
Cognitive Scientist

- Led multimillion-dollar NASA-funded project as Principal Investigator to evolve decision-support interfaces into bi-directional, transparent agent communication models, reducing workload by 29%.
- Directed user experience development program for AI-powered flight deck system, establishing design-to-implementation workflows through collaboration with UX designers on prototypes.
- Created reusable framework for human-autonomy interaction patterns used across programs to prototype and evaluate AI systems, including advanced flight deck and spacesuit platforms.
- Created an advanced air mobility agent that used reinforcement learning to adjust alerting preferences from a simulated pilot's feedback, enabling personalized, context-aware behavior.
- Mentored Iowa State University student as Program Manager for NSF project on playbook interfaces with AI agents.

Independent Contractor 2025-Present

- Utilized large language models to synthesize technical research and solutions for The International Rescue Committee, enabling renewable energy solutions to replace diesel-powered irrigation systems.
- Leveraged large language models to devise a novel ROI calculation framework for Lumina Foundation, identifying data sources to enable continuous tracking of educational credentials.
- Developed algorithms for representing complex scientific concepts through symbolic communication for METI International, with potential applications in AI communication and knowledge systems.

Senior Cognitive Scientist Alion Science and Technology 2006-2012

- Designed novel tool that automatically generates predictive operator agents from recorded user interactions for simulation-based performance evaluation.
- Created AI agents with new spatial reasoning capabilities, enabling them to interpret environmental context and trigger procedures dynamically in complex domains such as robotic control.
- Created an agent that learns UAV maneuvering strategies from pilot data, achieving human-level performance in simulation and demonstrating principles for generalizing to physical flight environments.

Research Psychologist NASA Ames Research Center 2000-2006

- Contributed to design and evaluation of shuttle interface that improved accuracy by 43%, reduced resolution time by 54%, and reduced workload by 37%.
- Supported development and field study of Mars Rover collaborative workspace.

Research Programmer Carnegie Mellon University 1989-1996

- Designed and administered experiments for cognitive psychology research.
- Contributed to computational models of human learning and decision making.

Patents

U.S. Patent 12,042,289 A1, "Cognitive Battery for Return to Service"	July 23, 2024
U.S. Patent 11,533,813 B1, "Touchscreen Display Gripping Mechanism"	December 20, 2022
U.S. Patent 11,004,347 B1, "Looming Ball Pilot Guidance and Cuing Symbology"	May 11, 2021

Certifications

Human Subject Research Certification for U.S. Air Force, CITI ID 67443149	Feb 2023
UX Certification in Interaction Design, Nielsen Norman Group, ID 1046162	May 2021
Small UAS Remote Pilot Certification, Federal Aviation Administration, ID 4262186	May 2019

Awards

Collins Aerospace Achievement Award - NASA Project Innovation, 2025
Collins Aerospace Spot Award - Collaboration, 2024
Collins Aerospace Spot Award - Advanced Technology Innovation, 2023
Collins Aerospace Spot Award - NASA engagement, 2023
Collins Aerospace Excellence Award - Exceptional performance on cross-SBU space suit project, 2019
NASA Software Release Award - Reduced Crew Operations (RCO) ground station, 2016
NASA Group Achievement Award - Conducting simulation evaluations with the NASA Ames Single Pilot Operations Experiment Team, 2014
NASA Group Achievement Award - For exemplary performance and cohesive teamwork in achieving NASA missions and objectives, 2013
Recommended Reading List Author - Behavior Representation in Modeling and Simulation (BRIMS), 2004
NASA Turning Goals into Reality (TGIR) Award - Human Factors contributor to the Ames MER Infusion Team, 2004
NASA Ames Honor Award - Member of Code I MER Technology Infusion Team, 2003
NASA Group Achievement Award - Successful collaboration with JPL infusing Ames technology into MER Mission, 2003

Publications and Presentations

Refereed journal articles

- Ganeriwala, P., Narayan, N., Jones, R. M., Matessa, M., Bhattacharyya, S., Davis, J., Rollini, S. F., Purohit, H., & Neogi, N. (2025). Systems Engineering with Architecture Modeling, Formal Verification and Human Interactions for Learning-Enabled Autonomous Agent. *Systems Engineering*, 28(5), 648-658.
- Matessa, M. (2024). Algorithms for encoding interstellar messages. *Acta Astronautica*, 229, 161-165.
- Matessa, M. (2022). Algorithms for decoding interstellar messages. *Journal of the British Interplanetary Society*, 75, 357-361.
- Matessa, M., Barkow, J. H., & DeVito, C. L. (2022). Other Messages for Other Senses. *Journal of the British Interplanetary Society*, 75, 362-367.
- Matessa, M., Vakoch, D. A., Lower, T. A., Barkow, J. H., & DeVito, C. L. (2022). How to decode interstellar messages. *Acta Astronautica*, 192, 77-81.
- Tokadlı, G., Dorneich, M., & Matessa, M. (2021). Evaluation of Playbook Delegation Approach in Human-Autonomy Teaming for Single Pilot Operations. *International Journal of Human-Computer Interaction*, 1044-7318.
- Tokadlı, G., Dorneich, M. C., & Matessa, M. (2021). Toward Human-Autonomy Teaming in Single-Pilot Operations: Domain Analysis and Requirements. *Journal of Air Transportation*, 29(4), 142-152.
- Lachter, J., Brandt, S. L., Battiste, V., Matessa, M., & Johnson, W. W. (2017). Enhanced ground support: lessons from work on reduced crew operations. *Cognition, Technology & Work*, 19(2-3), 279-288.
- Vakoch, D., & Matessa, M. (2011). An Algorithmic Approach to Communicating Reciprocal Altruism in Interstellar Messages. *Acta Astronautica*, 68(3-4), pp 459-475.
- Ball, J., Myers, C., Heiberg, A., Cooke, N., Matessa, M., Freiman, M., & Rogers, S. (2010). The Synthetic Teammate Project. *Computational and Mathematical Organization Theory*, 16(3), 271-299.
- Vera, A., John, B., Remington, R., Matessa, M., & Freed, M. (2005). Automating Human-Performance Modeling at the Millisecond Level. *Human-Computer Interaction*, 20 (3), 225-265.
- Matessa, M., & Anderson, J. R. (2000). Modelling Focused Learning in Role Assignment. *Language and Cognitive Processes*, 15 (3), 263-292.
- Anderson, J. R., Bothell, D., Lebiere, C. & Matessa, M. (1998). An Integrated Theory of List Memory. *Journal of Memory and Language*, 38, 341-380.
- Anderson, J. R., Matessa, M. & Lebiere, C. (1997). ACT-R: A Theory of Higher Level Cognition and Its Relation to Visual Attention. *Human-Computer Interaction*, 12, 439-462.
- Anderson, J. R. & Matessa, M. (1997). A production system theory of serial memory. *Psychological Review*, 104 (4), 728-748.
- Anderson, J. R. & Matessa, M. (1992). Explorations of an incremental, Bayesian algorithm for categorization. *Machine Learning*, 9, 275-308.

Refereed conference proceedings papers

- Ganeriwala, P., Matessa, M., Bhattacharyya, S., et al. (2025). Design and Validation of Learning Aware HMI For Learning-Enabled Increasingly Autonomous Systems. In *Proceedings of the 2025 IEEE International Systems Conference (SysCon)*, (pp. 1-8). IEEE.
- Matessa, M., & Rohrig, J. (2023, September). Human-Autonomy Teaming Guidelines with a Spacesuit Example. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* (Vol. 67, No. 1, pp. 477-482). Sage CA: Los Angeles, CA: SAGE Publications. doi:10.1177/21695067231192198
- Narayan, N., Ganeriwala, P., Jones, R. M., Matessa, M., Bhattacharyya, S., Davis, J., Purohit, H., & Rollini, S. F. (2023). Assuring Learning-Enabled Increasingly Autonomous Systems. In *2023 IEEE International Systems Conference (SysCon)* (pp. 1-7). IEEE.
- Bhattacharyya, S., Davis, J., Gupta, A., Narayan, N., & Matessa, M. (2021). Assuring Increasingly Autonomous Systems in Human-Machine Teams: An Urban Air Mobility Case Study. In Farrell M., Luckcuck M. (Eds.), *Third workshop on formal methods for autonomous systems (FMAS 2021)* (pp. 150–166, Vol. 348). *Electronic Proceedings in Theoretical Computer Science*. <https://doi.org/10.4204/EPTCS.348.11>

- Tokadlı, G., Dorneich, M. C., & Matessa, M. (2019, November). Development Approach of Playbook Interface for Human-Autonomy Teaming in Single Pilot Operations. In Proceedings of the Human Factors and Ergonomics Society Annual Meeting (Vol. 63, No. 1, pp. 357-361). Sage CA: Los Angeles, CA: SAGE Publications.
- Tokadlı, G., Dorneich, M. C., Matessa, M., & Eda, S. (2019, September). The Evaluation of a Playbook Interface for Human-Autonomy Teaming in Single Pilot Operations. In 2019 IEEE/AIAA 38th Digital Avionics Systems Conference (DASC) (pp. 1-7). IEEE.
- Carrico, M., Matessa, M., & Stover, K. (2018). Reduced Crew Operations. In 2018 IEEE/AIAA 37th Digital Avionics Systems Conference (DASC) (pp. 1-8). IEEE.
- Matessa, M., Vu, K., Strybel, T., Battiste, V., Schnell, T., & Cover, M. (2018). Using Distributed Simulation to Investigate Human-Autonomy Teaming. Proceedings of the 20th International Conference on Human-Computer Interaction (pp 541-550)..
- Cover, M., Reichlen, C., Matessa, M., Schnell, T. (2018). Analysis of Airline Pilots Subjective Feedback to Human Autonomy Teaming in a Reduced Crew Environment. Proceedings of the 20th International Conference on Human-Computer Interaction.
- Matessa, M. (2017). Using a crew resource management framework to develop human-autonomy teaming measures. In the International Conference on Applied Human Factors and Ergonomics (pp. 46-57). Springer, Cham.
- Matessa, M. (2017). Using a Crew Resource Management Framework to Develop Human-Autonomy Teaming Measures. AHFE 2017.
- Shively, R., Lachter, J., Brandt, S., Matessa, M., Battiste, V., & Johnson, W. (2017). Why Human-Autonomy Teaming? AHFE 2017.
- Shively, R. J., Brandt, S., Lachter, J., Matessa, M., Sadler, G., Battiste, H. (2016). Application of Human-Autonomy Teaming (HAT) Patterns to Reduced Crew Operations (RCO). Proceedings of the 18th International Conference on Human-Computer Interaction.
- Shively, R. J., Brandt, S. L., Lachter, J., Matessa, M., Sadler, G., & Battiste, H. (2016, July). Application of Human-Autonomy Teaming (HAT) Patterns to Reduced Crew Operations (RCO). In the International Conference on Engineering Psychology and Cognitive Ergonomics (pp. 244-255). Springer, Cham.
- Bhattacharyya, S., Davis, J., Vogl, T., Fix, M., Mclean, A., Matessa, M., Smith-Velazquez, L. (2015). Enhancing Autonomy with Trusted Cognitive Modeling. AUVSI 2015. Atlanta, GA.
- Dao, Q., Koltai, K., Cals, S., Brandt, S., Lachter, J., Matessa, M., Smith, D., Battiste, V. & Johnson, W. (2015). Evaluation of a recommender system for single-pilot operations, Proceedings of the 6th International Conference on Applied Human Factors and Ergonomics, Las Vegas, NV.
- Ligda, S. V., Fischer, U., Mosier, K., Matessa, M., Battiste, V., & Johnson, W. W. (2015). Effectiveness of Advanced Collaboration Tools on Crew Communication in Reduced Crew Operations. Proceedings of the 17th International Conference on Human-Computer Interaction, Los Angeles, CA.
- Ligda, S. V., Fischer, U., Mosier, K., Matessa, M., Battiste, V., & Johnson, W. W. (2015, August). Effectiveness of advanced collaboration tools on crew communication in reduced crew operations. In International conference on engineering psychology and cognitive ergonomics (pp. 416-427). Springer, Cham.
- Matessa, M., Strybel, T., Vu, K., Battiste, V., Schnell, T. (2015). Concept of Operations for RCO/SPO.
- Lachter, J., Battiste, V., Matessa, M. and Johnson, W.W. (2014). Toward Single Pilot Operations: Impact of the loss of non-verbal communication on the flight deck. *Proceedings of the HCI-AERO 2014 conference, Silicon Valley, California, July 30-August 1, 2014*
- Lachter, J., Brandt, S., Battiste, V., Ligda, S., Matessa, M. and Johnson, W.W. (2014). Toward Single Pilot Operations: Developing a Ground Station. *Proceedings of the HCI-AERO 2014 conference, Silicon Valley, California, July 30-August 1, 2014*
- Foltz, P. W., Sparks, R., Rosenstein, M., Carley, K. M., Martin, K. M., Warwick, W., Matessa, M., LaVoie N. and Lauper, U (2009). Developing an Integrated Toolset for the Tactical Human Integration of Networked Knowledge ATO, In proceedings of the 4th Annual West Point Network Science Workshop, West Point, NY.
- Matessa, M., Archer, R., & Mui, R. (2007). Dynamic Spatial Reasoning Capability in a Graphical Interface Evaluation Tool. Proceedings of the Eighth International Conference on Cognitive Modeling (ICCM '07), Ann Arbor, MI.
- Archer, S., Archer, R., & Matessa, M. (2007). Our GRBIL has a Split Personality. Proceedings of the SAE

- Digital Modeling Conference, June 2007, Seattle, WA.
- Matessa, M. & Brockett, A. (2007). Using a Diagram Representation System with ACT-R. *Proceedings of the Sixteenth Conference on Behavior Representation in Modeling and Simulation* (BRIMS '07), Norfolk, VA.
- Matessa, M., & Polson, P. (2006). List Models of Procedure Learning. *Proceedings of the International Conference on Human-Computer Interaction in Aeronautics (HCI-Aero)*, San Francisco, CA
- Matessa, M., & Ravinder, U. (2006). Modeling Multitasking with Templates of Human Performance in Space Shuttle Procedures. *Habitation*, 10 (3/4), 201-202.
- Matessa, M., & Remington, R. (2005, October). Reusable templates of human performance in space shuttle procedures. In *2005 IEEE International Conference on Systems, Man and Cybernetics* (Vol. 2, pp. 1610-1614). IEEE.
- Matessa, M., & Remington, R. (2005, September). Eye movements in human performance modeling of space shuttle operations. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* (Vol. 49, No. 12, pp. 1114-1118). Sage CA: Los Angeles, CA: SAGE Publications.
- Huemer, V., Matessa, M., & McCann, R. (2005). Fault Management during Dynamic Phases of Spacecraft Flight: Effects of Cockpit Display Format and Workload. To appear in *Proceedings of IEEE Systems, Man, and Cybernetics Conference*, Waikoloa, HI.
- Matessa, M. (2004). An ACT-R Framework for Interleaving Templates of Human Behavior. *Proceedings of the Twenty-sixth Annual Conference of the Cognitive Science Society* (pp. 903-908), Chicago, IL.
- Matessa, M. (2004). Anticipatory Eye Movements in Interleaving Templates of Human Behavior. *Proceedings of the Sixth International Conference on Cognitive Modeling* (pp. 184-189), Pittsburgh, PA.
- Remington, R.W., Lee, S. M., Ravinder, U., & Matessa, M (2004). Observations on Human Performance in Air Traffic Control Operations: Preliminaries to a Cognitive Model. *Proceedings of Behavior Representation in Modeling and Simulation* (BRIMS '04), Arlington, VA.
- Lee, S. M., Remington, R.W., Ravinder, U., & Matessa, M (2004). Developing human performance models using Apex/CPM-GOMS for agent-based modeling and simulation. *Proceedings of the Military, Government, and Aerospace Simulation Symposium of the Advanced Simulation Technologies Conference* (pp. 49-54), Arlington, VA.
- Remington, R., Matessa, M., Freed, M., & Lee, S.M. (2003). Using Apex / CPM-GOMS to Develop Human-Like Software Agents. *Proceedings of the Second International Joint Conference on Autonomous Agents and Multiagent Systems*, Melbourne, Australia.
- Matessa, M. (2003). Measures of Adaptive Communication. *Proceedings of the Second Workshop on Empirical Evaluation of Adaptive Systems* (pp. 41-50), Johnstown, PA.
- Freed, M., Matessa, M., Remington, R., Vera, A. (2003). How Apex Automates CPM-GOMS. *Proceedings of the Fifth International Conference on Cognitive Modeling* (pp. 93-98), Bamberg, Germany.
- John, B., Vera, A., Matessa, M., Freed, M, & Remington, R. (2002). Automating CPM-GOMS. *Proceedings of CHI* (pp. 147-154). ACM, New York, NY.
- Matessa, M. (2001). The Benefit of Structured Interfaces in Collaborative Communication. In *Working Notes of the 2001 AAAI Fall Symposium on Intent Inference for Collaborative Tasks* (pp. 62-64), Cape Cod, MA.
- Matessa, M. (2001). Interactive Models of Collaborative Communication. *Proceedings of the Twenty-third Annual Conference of the Cognitive Science Society* (pp. 634-638). Hillsdale, NJ: Lawrence Erlbaum Associates.
- Matessa, M., & Anderson J. R. (2000). An ACT-R Model of Adaptive Communication. *Proceedings of the Third International Conference on Cognitive Modeling* (pp. 210-217). Groningen, NL: Universal Press.
- Matessa, M., & Anderson J. R. (1999). Towards an ACT-R Model of Communication in Problem Solving. In *Working Notes of the 1999 AAAI Fall Symposium on Psychological Models of Communication in Collaborative Systems* (pp. 67-72), Cape Cod, MA.
- Byrne, M. D., Anderson, J. R., Douglass, S., & Matessa, M. (1999). Eye tracking the visual search of click-down menus. *Human Factors in Computing Systems: Proceedings of CHI 99* (pp. 402-409). Reading, MA: Addison Wesley.
- Anderson, J. R., Matessa, M. & Douglass, S. (1995). The ACT-R theory and visual attention. *Proceedings of the Seventeenth Annual Conference of the Cognitive Science Society* (pp. 61-65). Hillsdale, NJ: Lawrence Erlbaum Associates.
- Lee, F. J., Anderson, J. R. & Matessa, M. P. (1995). Components of Dynamic Skill Acquisition. *Proceedings of the Seventeenth Annual Conference of the Cognitive Science Society* (pp. 506-511). Hillsdale, NJ:

Lawrence Erlbaum Associates.

- Ward, W., Issar, S. Huang, X., Hon, H., Hwang, M., Young, S., Matessa, M., Liu, F., Stern, R. (1992). Speech Understanding In Open Tasks. Proceedings of the Fifth DARPA Workshop on Speech and Natural Language, February 1992.
- Young, S., & Matessa, M. (1991). Using Pragmatic and Semantic Knowledge to Correct Parsing of Spoken Language Utterances. Proceedings of the 2nd European Conference on Speech Communication and Technology (Eurospeech), (pp. 223-227). Genova, Italy, September 1991.
- Anderson, J. R., & Matessa, M. (1990). A rational analysis of categorization. Proceedings of the Seventh International Machine Learning Conference (pp. 76-84). Palo Alto, CA: Morgan Kaufman.

Book chapters

- Matessa, M. (2017). Using a Crew Resource Management Framework to Develop Human-Autonomy Teaming Measures. In Hale, K. & Stanney, K. (Eds) *Advances in Neuroergonomics and Cognitive Engineering* (pp 46-57). Springer.
- Laughery, K. R., Plott, B., Matessa, M., Archer, S., & Lebiere, C. (2012). Modeling human performance in complex systems. In Salvendy, G. (Ed.) *2012 Handbook of Human Factors and Ergonomics* (pp 931-961). John Wiley & Sons.
- Matessa, M. & Warwick, W. (2011). Virtual and Constructive Simulations with the GRBIL Modeling Tool. In L. Rothrock & S. Narayanan (Eds.), *Human-in-the-Loop Simulations*. Springer.
- Matessa, M., & Mui, R. (2009). GRBIL: A Tool for Dynamic Operator and Environment Modeling. In P. McDermott, & L. Allender (Eds.), *Advanced Decision Architectures for the Warfighter: Foundations and Technology* (pp 237-254). Army Research Laboratory.
- Anderson, J. R. & Matessa, M. (1998). The rational analysis of categorization and the ACT-R architecture. In M. Oaksford & N. Chater (Eds.), *Rational Models of Cognition*. Oxford University Press.
- Anderson, J. R., Matessa, M., & Lebiere, C. (1998). The Visual Interface. In J. R. Anderson & C. Lebiere (Eds.), *The Atomic Components of Thought*. Mahwah, NJ: Erlbaum.
- Anderson, J. R., Bothell, D., Lebiere, C. & Matessa, M. (1998). List Memory. In J. R. Anderson & C. Lebiere (Eds.), *The Atomic Components of Thought*. Mahwah, NJ: Erlbaum.
- Anderson, J. R. & Matessa, M. (1991). An iterative Bayesian algorithm for categorization. In D. Fisher, M. J. Pazzani & P. Langley (Eds.), *Concept Formation: Knowledge and Experience in Unsupervised Learning*. San Mateo, CA: Morgan Kaufman.

Posters/presentations

- Matessa, M. (2026). Measuring Dynamic Trust. NSF Human-AI Complementarity Workshop. Pittsburgh, PA.
- Dorneich, M., Tokadli, G., Matessa, M., & Shapiro, G. (2018). Development of Playbook Interaction Framework for Reduced Crew Operations. Industrial Advisory Board (IAB) Meeting. Tewksbury, MA.
- Vakoch, D., Matessa, M., DeVito, C., & Kaiser, A. (2018). A mathematical and scientific message for active SETI. Presentation given at 42nd COSPAR Scientific Assembly, 14-22 July 2018, in Pasadena, California, USA.
- Matessa (2017). Human-Autonomy Teaming Measures. Presentation given at Autonomy and the Next Generation Flight Deck Symposium. NASA/SAE. Moffett Field, CA.
- Matessa (2015). Reduced Crew Operations Research. Presentation given at Air Line Pilots Association (ALPA) Conference. Washington, DC.
- Matessa (2014). Issues in Nominal and Off-nominal SPO. Presentation given at HCI-Aero. Santa Clara, CA.
- Warwick, W., Carley, K., Martin, M., & Matessa, M. (2011) Integrating Agent-based Simulation and Social Network Analysis. Talk presented at the Human Social Culture Behavior Modeling conference, Chantilly, VA. February 8-10.
- Matessa, M. (2008). Spatial Reasoning in Cognitive Architectures. Poster presented at the Thirtieth Annual Conference of the Cognitive Science Society, Washington, DC.
- Archer, R., & Matessa, M. (2007). User Interface Evaluation and Development using the GRaph Based Interface Language (GRBIL) Tool. Tutorial Presented at the Sixteenth Conference on Behavior

- Representation in Modeling and Simulation, Norfolk, VA.
- Archer, R., Mui, R. & Matessa, M. (2007). Integrating IMPRINT and ACT-R. ADA RMB Meeting, February 21-23. Westminster, CO.
- Matessa, M. (2006). Four levels of communication, error, and recovery in ACT-R. Paper presented at the Thirteenth Annual ACT-R Workshop. Carnegie Mellon University, July 2006.
- John, B. E., Lewis, R. L., Matessa, M., Remington, R., Vera, A. (2005). Scalable Cognitive Modeling Through Compositional Reuse. Office of Naval Research Conference on Affordable Human Behavior Modeling, Arlington, VA. June, 2005.
- Matessa, M. & Remington, R. (2005). Model-based Support for Crew Activity Planning. Bioastronautics Investigators' Workshop, Galveston, TX. January, 2005.
- Hayashi, M., Huemer, V., Renema, F., Elkins, S., Lawrence, B., Matessa, M., McCandless, J. & McCann, R. (2005). Effects of the Space Shuttle Cockpit Avionics Upgrade on Crews' Performance and Situation Awareness. Bioastronautics Investigators' Workshop, Galveston, TX. January, 2005.
- Matessa, M. (2004). ACT-Stitch: Complex Predictions from Simple Descriptions. Human Factors Symposium, NASA Ames Research Center, Moffett Field, CA.
- John, B. E., Lewis, R. L., Matessa, M., Remington, R., Vera, A. (2004). Scalable Cognitive Modeling Through Compositional Reuse. Office of Naval Research Conference on Affordable Human Behavior Modeling, Arlington, VA. July, 2004.
- Remington, R., Matessa, M., Freed, M., Lee, S., & Ravinder, U. (2003). Scalable Human Performance Modeling. Paper presented at Human Performance Modeling Workshop. NASA Ames Research Center. October, 2003.
- Matessa, M. & Polson, P. (2003). Re-use of a serial list model for aircraft procedure learning. Paper presented at the Tenth Annual ACT-R Workshop. Carnegie Mellon University, July 2003.
- Remington, John, Vera, Matessa, Freed, Dalal, Harris, & Dahlman. (2002). Apex/CPM-GOMS. Cognitive Science Society Conference Tutorial, Fairfax, VA.
- Matessa, M., Vera, A., John, B., Remington, R., & Freed, M. (2002). Reusable Templates in Human Performance Modeling. Proceedings of the Twenty-fourth Annual Conference of the Cognitive Science Society (pp. 649-654). Mahwah, NJ: Erlbaum.
- Matessa, M. (2002). Mathematical Descriptions of Human Behavior for Interstellar Communication. The Art and Science of Interstellar Communication Workshop. Paris, March 2002.
- Matessa, M., & Anderson, J.R. (1999). Modeling Collaborative Communication. Poster presented at the Twenty First Annual Conference of the Cognitive Science Society.
- Matessa, M., & Anderson, J. R. (1999). Modeling communication in a graph comparison task. Paper presented at the Sixth Annual ACT-R Workshop. George Mason University, August 1999.
- Matessa, M., & Anderson, J.R. (1998). Testing a Model of Role Assignment. Proceedings of the Twentieth Annual Conference of the Cognitive Science Society. Hillsdale, NJ: Lawrence Erlbaum Associates.
- Matessa, M., & Anderson, J.R. (1997). Focused Learning in a Linguistic Environment. Proceedings of the Nineteenth Annual Conference of the Cognitive Science Society. Hillsdale, NJ: Lawrence Erlbaum Associates.

Technical reports

- Davis, J., Matessa, M., Rollini, S. F., Bhattacharyya, S., Gupta, A., Narayan, N., Ganeriwala, P., Purohit, H., & Jones, R. M. (2023). Assured Human Machine Interface for Increasingly Autonomous Systems (AHMIAS). NASA Contractor Report.
- Matessa, M., & Lovett, M. (2008). Optimal Training Systems. Technical Report AFRL-SR-AR-TR-08-0258, Air Force Research Laboratory.
- Beutter, B. R., Matessa, M., McCandless, J. W., McCann, R. S., Spirkovska, L., Liston, D., Hayashi, M., Huemer, V., Lachter, J., Ravinder, U., Elkins, S., Renema, F., Lawrence, R., & Hamilton, A. (2006). Evaluation of an Onboard Real-Time Fault Management Support System for Next-Generation Space Vehicles (Intelligent Spacecraft Interface Systems Lab Report). Moffett Field, CA: NASA Ames Research Center.
- Matessa, M., & Polson, P. (2005). List Models of Procedure Learning. NASA Technical Memorandum

2005-213465. Ames Research Center, Moffett Field, CA.

Vera, A., Matessa, M., Freed, M., & Remington, R. (2004). Using Apex To Construct CPM-GOMS Models. NASA Tech Brief ARC-15072-1. Ames Research Center, Moffett Field, CA.

Matessa, M. (2004). An ACT-R Framework for Interleaving Templates of Human Behavior. NASA-TM 2004-212838. Ames Research Center, Moffett Field, CA.

Matessa, M. (2004). Anticipatory Eye Movements in Interleaving Templates of Human Behavior. NASA-TM 2004-212839. Ames Research Center, Moffett Field, CA.

Vera, A., Remington, R., Matessa, M., & Freed, M. (2001). Facilitating Human Performance Modeling. NASA Research And Technology Report. Ames Research Center, Moffett Field, CA.

Matessa, M. (1996). The ACT-R Visual Interface Manual. Psychology Department, Carnegie Mellon University, Pittsburgh, PA.

Young, Sheryl R. & Matessa, M. (1992). Using Pragmatic and Semantic Knowledge to Correct Parsing of Spoken Language Utterances. Technical report CMU-CS-92-120, Computer Science Department, Carnegie Mellon University, Pittsburgh, PA.

Young, Sheryl R. & Matessa, M. (1992). MINDS-II Feedback Architecture: Detection and Correction of Speech Misrecognitions. Technical report CMU-CS-92-119, Computer Science Department, Carnegie Mellon University, Pittsburgh, PA.

Languages/Tools

Lisp (proficient), Python (NumPy and pandas for data analysis)

Invited talks

Human-AI Teaming in Aerospace and Energy. Invited talk given to Pacific Northwest National Laboratory (PNNL), Richland, WA, 2025.

Human-Autonomy Teaming Guidelines for Spacesuits. Invited talk given at the University of Michigan NASA SUITS Hackathon, Ann Arbor, MI, 2024

Extraterrestrial Messages. Invited talk given at the California Academy of Sciences, San Francisco, CA, 2018

Creating Messages for Extraterrestrials. Invited talk given at METI International, San Francisco CA, 2018

Single Pilot Operations/Reduced Crew Operations. Invited talk given at the University of Iowa, Iowa City, IA, 2015

Analysis of Accommodation. Presented at Communication Analysis Workshop. Tempe, AZ, 2010

GRBIL and ACT-R. Presented at Carnegie Mellon University, Pittsburgh, PA, 2009

Introduction to ACT-R. Invited talk given at Stanford University, Palo Alto, CA, 2004

Adaptive Communication and Compositional Modeling. Invited talk given at Rensselaer Polytechnic Institute, Troy, NY, 2003

New Developments in the ACT-R Visual Interface. Invited talk given at the University of Saarbruecken, Germany, 1995

Professional Memberships

BayCHI

Other

SLV Unified School District AI Task Force Member

Volunteer Disaster Service Worker

Citizens' Climate Lobby Volunteer

Citizen Science Volunteer (Planet Hunters, Solar Stormwatch, Galaxy Zoo, Citizen Sky)

Steering Committee Member of local Transition Town, 2010-2016

Community Facilitator for MIT's Climate CoLab collaborative workspace, 2011-2013

Six months of private pilot training, including flying solo

NAUI Scuba certified

Technician class amateur radio license (KB3TCT)

Amateur variable star observer (MMIK)